CHAPTER 19
SOIL–STRUCTURE INTERACTION FOR SEISMIC DESIGN

19.1 GENERAL

19.1.1 Scope. Determination of the design earthquake forces and the corresponding displacements of the structure is permitted to consider the effects of soil–structure interaction (SSI) in accordance with this section. SSI may be used in conjunction with the equivalent lateral force procedure of Section 19.2.1, linear dynamic analysis procedures of Section 19.2.2, or the nonlinear response history procedure of Section 19.2.3 when the structure is located on Site Class C, D, E, or F. When soil–structure interaction effects are considered, the analytical model of the structure shall directly incorporate horizontal, vertical, and rotational foundation and soil flexibility. For the purpose of this section, both upper and lower bound estimates for the foundation and soil stiffnesses per Section 12.13.3 shall be considered. The case that results in the lesser reduction or greater amplification in response parameters shall be used for design.

If the provisions of this chapter are used, then Section 12.8.1.3 shall not apply.

19.1.2 Definitions. The following definitions apply to the provisions of Chapter 19 and are in addition to the definitions presented in Chapter 11:

BASE SLAB AVERAGING: Kinematic SSI of a shallow (nonembedded) foundation caused by wave incongruence over the base area.

FOUNDATION INPUT MOTION: Motion that effectively excites the structure and its foundation.

FREE-FIELD MOTION: Motion at ground surface in absence of structure and its foundation.

INERTIAL SSI: The dynamic interaction between the structure, its foundation, and the surrounding soil caused by the foundation input motion.

KINEMATIC SSI: The modification of free-field ground motion caused by nonvertical incident seismic waves and spatial incoherence; the modification yields the foundation input motion.

RADIATION DAMPING: The damping in the soil–structure system caused by the generation and propagation of waves away from the foundation, which are caused by dynamic displacements of the foundation relative to the free-field displacements.

SOIL DAMPING: The hysteretic (material) damping of the soil.

19.1.3 Symbols. The following symbols apply only to the provisions of Chapter 19 as indicated and are in addition to the symbols presented in Chapter 11:

a_o = dimensionless frequency [Eqs. (19.3-11) and (19.3-21)]

B = half the smaller dimension of the base of the structure

B_{SSI} = the factor to adjust the design response spectrum and MCE_R response spectrum in accordance with Sections 11.4.6 and 11.4.7 or a site-specific response spectrum for damping ratios other than 0.05 [Eq. (19.2-4)]

b_e = effective foundation size [Eqs. (19.4-4) and (19.4-4si)]

$\tilde{C}_s$ = the seismic response coefficient determined in accordance with Section 12.8.1.1 assuming a flexible structural base at the foundation–soil interface in accordance with Section 19.1

D_s = the depth of a soft layer overlaying a stiff layer [Eq. (19.3-4)]

e = foundation embedment depth

G = effective shear modulus derived or approximated based on G_0 and Table 19.3-2

G_0 = the average shear modulus for the soils beneath the foundation at small strain levels

h^* = effective structure height

K_{xx}, K_{rr} = rotational foundation stiffness [Eqs. (19.3-9) and (19.3-19)]

K_y, K_r = translational foundational stiffness [Eqs. (19.3-8) and (19.3-18)]

L = half the larger dimension of the base of the structure

M^* = effective modal mass for the fundamental mode of vibration in the direction under consideration

RRS_{bsa} = site-specific response spectral modification factor for base-slab averaging [Eq. (19.4-1)]

RRS_e = site-specific response spectral modification factor for foundation embedment [Eq. (19.4-5)]

r_f = radius of the circular foundation

$\tilde{S}_a$ = response spectral acceleration including the effects of SSI [Eqs. (19.2-5) through (19.2-8)]

T = fundamental period of the structure determined in accordance with Section 12.8.2 based on a mathematical model with a fixed base condition. The upper bound limitation of C_uT_a on the fundamental period from Section 12.8.2 shall not apply, and the approximate structural period, T_a, shall not be used

$\tilde{T}$ = fundamental period of the structure using a model with a flexible base in accordance with Section 19.1.1. The upper bound limitation of C_uT_a on the fundamental period from Section 12.8.2 shall not apply, and the approximate structural period, T_a, shall not be used

$(\tilde{T}/T)_{\text{eff}}$ = effective period lengthening that depends on expected ductility demand, μ [Eq. (19.3-2)]

T_{xx}, T_{rr} = fundamental translational period of SSI system [Eqs. (19.3-7) and (19.3-17)]

T_y, T_r = fundamental translational period of SSI system [Eqs. (19.3-6) and (19.3-16)]

$\tilde{V}$ = base shear adjusted for soil–structure interaction [Eq. (19.2-1)]

$\tilde{V}_t$ = base shear adjusted for soil–structure interaction determined through modal response spectrum analysis

v_s = the average effective shear wave velocity over a depth specified in Section 19.3 and 19.4 specific to the SSI effect being computed determined using v_{so} and Table 19.3-1 or a site-specific geotechnical study

v_{so} = the average shear wave velocity for the soils beneath the foundation at small strain levels (10^{-3}% or less) over a depth specified in Sections 19.3 and 19.4 specific to the SSI effect being computed

$\bar{W}$ = weight caused by the modal mass in the fundamental mode, or alternatively, the effective seismic weight in accordance with Section 12.7.2

α = coefficient that accounts for the reduction in base shear caused by foundation damping SSI

α_{xx}, α_{rr} = dimensionless factor, function of dimensionless frequency, a_0 [Eqs. (19.3-14) and (19.3-24)]

β = effective viscous damping ratio of the structure, taken as 5% unless otherwise justified by analysis

β_0 = the effective viscous damping ratio of the soil–structure system, based on Section 19.3.2 [Eq. (19.3-1)]

β_f = effective viscous damping ratio relating to foundation–soil interaction [Eq. (19.3-3)]

β_{rd} = radiation damping ratio determined in accordance with Section 19.3.3 or 19.3.4 [Eqs. (19.3-5) and (19.3-15)]

β_s, β_s' = soil hysteretic damping ratio determined in accordance with Section 19.3.5

β_{xx}, β_{rr} = rotational foundation damping coefficient [Eqs. (19.3-12) and (19.3-22)]

β_y, β_r = translational foundation damping coefficient [Eqs. (19.3-10) and (19.3-20)]

γ = the average unit weight of the soils over a depth of B below the base of the structure

μ = expected ductility demand

ν = Poisson's ratio; it is permitted to use 0.3 for sand and 0.45 for clay soils

Ψ = dimensionless factor, function of Poisson's ratio [Eqs. (19.3-13) and (19.3-23)]

19.2 SSI ADJUSTED STRUCTURAL DEMANDS

19.2.1 Equivalent Lateral Force Procedure. To account for the effects of SSI using a linear static procedure, the base shear, V, determined from Eq. (12.8-1) is permitted to be modified as follows:

$$\tilde{V} = V - \Delta V \geq \alpha V \quad (19.2\text{-}1)$$

$$\Delta V = \left(C_s - \frac{\tilde{C}_s}{B_{\text{SSI}}}\right)\bar{W} \quad (19.2\text{-}2)$$

$$\alpha = \begin{cases} 0.7 & \text{for } R \leq 3 \\ 0.5 + R/15 & \text{for } 3 < R < 6 \\ 0.9 & \text{for } R \geq 6 \end{cases} \quad (19.2\text{-}3)$$

$$B_{\text{SSI}} = 4/[5.6 - \ln(100\beta_0)] \quad (19.2\text{-}4)$$

where

$\tilde{V}$ = base shear adjusted for SSI;

V = the fixed-base structure base shear computed in accordance with Section 12.8.1;

R = the response modification factor in Table 12.2-1;

C_s = the seismic response coefficient determined in accordance with Section 12.8.1.1 assuming a fixed structural base at the foundation–soil interface;

$\tilde{C}_s$ = the seismic response coefficient determined in accordance with Section 12.8.1.1 assuming flexibility of the structural base at the foundation–soil interface in accordance with Section 19.1.1 using $\tilde{T}$ as the fundamental period of the structure in lieu of the fundamental period of the structure, T, as determined by Section 12.8.2;

$\bar{W}$ = weight caused by the effective modal mass in the fundamental mode, alternatively taken as the effective seismic weight in accordance with Section 12.7.2;

α = coefficient that accounts for the reduction in base shear caused by foundation damping SSI; and

β_0 = the effective viscous damping ratio of the soil–structure system, in accordance with Section 19.3.2.

The inclusion of kinematic interaction effects in accordance with Section 19.4 or other methods is not permitted with the equivalent lateral force procedure.

19.2.2 Linear Dynamic Analysis. To account for the effects of SSI, a linear dynamic analysis is permitted to be performed in accordance with Section 12.9 using either the SSI modified design response spectrum and MCE_R response spectrum in accordance with Sections 11.4.6 and 11.4.7 or SSI modified site-specific response spectrum, per Section 19.2.2.1 or an SSI modified site-specific response spectrum in accordance with Section 19.2.2.2 for spectral response acceleration, $\tilde{S}_a$, versus structural period, T. The resulting response spectral acceleration shall be divided by R/I_e, where I_e is prescribed in Section 11.5.1. The mathematical model used for the linear dynamic analysis shall include flexibility of the foundation and underlying soil in accordance with Section 19.1.1.

The inclusion of kinematic interaction effects in accordance with Section 19.4 or other methods is not permitted with a linear dynamic analysis procedure.

Scaling of the lateral forces from the modal response analysis shall be in accordance with Section 12.9.1.4 with calculated base shear, V, replaced with SSI adjusted base shear, $\tilde{V}$, determined in accordance with Eq. (19.2-1) and the modal base shear, V_t, replaced by the modal base shear calculated with the effects of SSI, $\tilde{V}_t$.

The modal base shear calculated with the effects of SSI, $\tilde{V}_t$, shall not be less than αV_t, where α is defined in Eq. (19.2-3).

19.2.2.1 SSI Modified General Design Response Spectrum. The general design response spectrum, which includes the effects of SSI to be used with the modal analysis procedure in Section 19.2.2, shall be developed as follows:

$$\tilde{S}_a = \left[\left(\frac{5}{B_{SSI}} - 2\right) \times \frac{T}{T_S} + 0.4\right] \times S_{DS} \quad (19.2\text{-}5)$$

for $0 < T < T_0$, and

$$\tilde{S}_a = S_{DS}/B_{SSI} \quad \text{for } T_0 \leq T \leq T_S, \quad \text{and} \quad (19.2\text{-}6)$$

$$\tilde{S}_a = S_{D1}/(B_{SSI}T), \quad \text{for } T_S < T \leq T_L, \quad \text{and} \quad (19.2\text{-}7)$$

$$\tilde{S}_a = S_{D1}T_L/(B_{SSI}T^2), \quad \text{for } T > T_L \quad (19.2\text{-}8)$$

where S_{DS} and S_{D1} are defined in Section 11.4.5; T_S, T_0, and T_L are as defined in Section 11.4.6; T is the period at the response spectrum ordinate; and B_{SSI} is defined in Eq. (19.2-4).

19.2.2.2 SSI Site-Specific Response Spectrum. A site-specific response spectrum that incorporates modifications caused by SSI is permitted to be developed in accordance with the requirements of Chapter 21. The spectrum is permitted to be adjusted for the effective viscous damping ratio of β_0, of the soil–structure system, as defined in Section 19.3.2, in the development of the site-specific spectrum.

19.2.3 Nonlinear Response History Procedure. It is permitted to account for the effects of SSI using a nonlinear response history analysis performed in accordance with Chapter 16 using acceleration histories scaled to a site-specific response spectrum modified for kinematic interaction in accordance with Section 19.4 or other approved methods. The mathematical model used for the analysis shall include foundation and soil flexibility per Section 19.1.1 and shall explicitly incorporate the effects of foundation damping per Section 19.3 or by other approved methods. Kinematic interaction effects per Section 19.4 are permitted to be included in the determination of the site-specific response spectrum.

The site-specific response spectrum shall be developed per the requirements of Chapter 21 with the following additional requirements:

1. The spectrum is permitted to be adjusted for kinematic SSI effects by multiplying the spectral acceleration ordinate at each period by the corresponding response spectrum ratios for either base slab averaging or embedment or both base slab averaging and embedment ($RRS_{bsa} \times RRS_e$) per Section 19.4, or by directly incorporating one or both of these effects into the development of the spectrum.
2. For structures embedded in the ground, the site-specific response spectrum is permitted to be developed at the depth of the embedded base level in lieu of at grade. For this case, the response spectrum ratio for embedment effects (RRS_e) shall be taken as 1.0.
3. The site-specific response spectrum modified for kinematic interaction shall not be taken as less than 80% of S_a as determined from a site-specific response spectrum in accordance with Section 21.3.
4. The site-specific response spectrum modified for kinematic interaction shall not be taken as less than 70% of S_a as determined from the design response spectrum and MCE_R response spectrum in accordance with Sections 11.4.6 and 11.4.7.

19.3 FOUNDATION DAMPING EFFECTS

19.3.1 Foundation Damping Requirements. Foundation damping effects are permitted to be considered through direct incorporation of soil hysteretic damping and radiation damping in the mathematical model of the structure.

The procedures of this section shall not be used for the following cases:

1. A foundation system consisting of discrete footings that are not interconnected and that are spaced less than the larger dimension of the supported lateral force-resisting element in the direction under consideration.
2. A foundation system consisting of, or including, deep foundations such as piles or piers.
3. A foundation system consisting of structural mats interconnected by concrete slabs that are characterized as flexible in accordance with Section 12.3.1.3 or that are not continuously connected to grade beams or other foundation elements.

19.3.2 Effective Damping Ratio. The effects of foundation damping shall be represented by the effective damping ratio of the soil–structure system, β_0, determined in accordance with Eq. (19.3-1):

$$\beta_0 = \beta_f + \frac{\beta}{(\tilde{T}/T)^2_{\text{eff}}} \le 0.20 \tag{19.3-1}$$

where

β_f = effective viscous damping ratio relating to foundation–soil interaction;
β = effective viscous damping ratio of the structure, taken as 5% unless otherwise justified by analysis; and
$(\tilde{T}/T)_{\text{eff}}$ = effective period lengthening ratio defined in Eq. (19.3-2).

The effective period lengthening ratio shall be determined in accordance with Eq. (19.3-2):

$$\left(\frac{\tilde{T}}{T}\right)_{\text{eff}} = \left\{1 + \frac{1}{\mu}\left[\left(\frac{\tilde{T}}{T}\right)^2 - 1\right]\right\}^{0.5} \tag{19.3-2}$$

where

μ = expected ductility demand. For equivalent lateral force or modal response spectrum analysis procedures, μ is the maximum base shear divided by the elastic base shear capacity; alternately, μ is permitted to be taken as R/Ω_0, where R and Ω_0 are per Table 12.2-1. For the response history analysis procedures, μ is the maximum displacement divided by the yield displacement of the structure measured at the highest point above grade.

The foundation damping ratio caused by soil hysteretic damping and radiation damping, β_f, is permitted to be determined in accordance with Eq. (19.3-3) or by other approved methods.

$$\beta_f = \left[\frac{(\tilde{T}/T)^2 - 1}{(\tilde{T}/T)^2}\right]\beta_s + \beta_{rd} \tag{19.3-3}$$

where

β_s = soil hysteretic damping ratio determined in accordance with Section 19.3.5, and
β_{rd} = radiation damping ratio determined in accordance with Section 19.3.3 or 19.3.4.

If a site more than a depth B or R below the base of the building consists of a relatively uniform layer of depth, D_s, overlaying a very stiff layer with a shear wave velocity more than twice that of the surface layer and $4D_s/v_s\tilde{T} < 1$, then the damping values, β_r, in Eq. (19.3-3) shall be replaced by β_s', per Eq. (19.3-4):

$$\beta_s' = \left(\frac{4D_s}{v_s\tilde{T}}\right)^4 \beta_s \tag{19.3-4}$$

19.3.3 Radiation Damping for Rectangular Foundations. The effects of radiation damping for structures with a rectangular foundation plan shall be represented by the effective damping ratio of the soil–structure system, β_r, determined in accordance with Eq. (19.3-5):

$$\beta_{\text{rd}} = \frac{1}{(\tilde{T}/T_y)^2}\beta_y + \frac{1}{(\tilde{T}/T_{xx})^2}\beta_{xx} \tag{19.3-5}$$

$$T_y = 2\pi\sqrt{\frac{M^*}{K_y}} \quad (19.3\text{-}6)$$

$$T_{xx} = 2\pi\sqrt{\frac{M^*(h^*)^2}{\alpha_{xx}K_{xx}}} \quad (19.3\text{-}7)$$

$$K_y = \frac{GB}{2-\nu}\left[6.8\left(\frac{L}{B}\right)^{0.65} + 0.8\left(\frac{L}{B}\right) + 1.6\right] \quad (19.3\text{-}8)$$

$$K_{xx} = \frac{GB^3}{1-\nu}\left[3.2\left(\frac{L}{B}\right) + 0.8\right] \quad (19.3\text{-}9)$$

$$\beta_y = \left[\frac{4(L/B)}{(K_y/GB)}\right]\left[\frac{a_0}{2}\right] \quad (19.3\text{-}10)$$

$$a_0 = \frac{2\pi B}{\tilde{T}v_s} \quad (19.3\text{-}11)$$

$$\beta_{xx} = \left[\frac{(4\psi/3)(L/B)a_0^2}{\left(\frac{K_{xx}}{GB^3}\right)\left[\left(2.2-\frac{0.4}{(L/B)^3}\right) + a_0^2\right]}\right]\left[\frac{a_0}{2\alpha_{xx}}\right] \quad (19.3\text{-}12)$$

$$\psi = \sqrt{\frac{2(1-\nu)}{(1-2\nu)}} \le 2.5 \quad (19.3\text{-}13)$$

$$\alpha_{xx} = 1.0 - \left[\frac{(0.55 + 0.01\sqrt{(L/B)-1})a_0^2}{\left(2.4-\frac{0.4}{(L/B)^3}\right) + a_0^2}\right] \quad (19.3\text{-}14)$$

where

M^* = effective modal mass for the fundamental mode of vibration in the direction under consideration;

h^* = effective structure height taken as the vertical distance from the foundation to the centroid of the first mode shape for multistory structures. Alternatively, h^* is permitted to be approximated as 70% of the total structure height for multistory structures or as the full height of the structure for one-story structures;

L = half the larger dimension of the base of the structure;

B = half the smaller dimension of the base of the structure;

v_s = the average effective shear wave velocity over a depth of B below the base of the structure determined using v_{so} and Table 19.3-1 or a site-specific study;

v_{so} = the average low strain shear wave velocity over a depth of B below the base of the structure;

G = effective shear modulus derived or approximated based on G_0 and Table 19.3-2;

$G_0 = \gamma v_{so}^2/g$, the average shear modulus for the soils beneath the foundation at small strain levels;

γ = the average unit weight of the soils over a depth of B below the base of the structure; and

ν = Poisson's ratio; it is permitted to use 0.3 for sandy and 0.45 for clayey soils.

19.3.4 Radiation Damping for Circular Foundations. The effects of radiation damping for structures with a circular foundation plan shall be represented by the effective damping ratio of the soil–structure system, β_{rd}, determined in accordance with (Eq. 19.3-15):

$$\beta_{rd} = \frac{1}{(\tilde{T}/T_r)^2}\beta_r + \frac{1}{(\tilde{T}/T_{rr})^2}\beta_{rr} \quad (19.3\text{-}15)$$

$$T_r = 2\pi\sqrt{\frac{M^*}{K_r}} \quad (19.3\text{-}16)$$

$$T_{rr} = 2\pi\sqrt{\frac{M^*(h^*)^2}{\alpha_{rr}K_{rr}}} \quad (19.3\text{-}17)$$

$$K_r = \frac{8Gr_f}{2-\nu} \quad (19.3\text{-}18)$$

$$K_{rr} = \frac{8Gr_f{}^3}{3(1-\nu)} \quad (19.3\text{-}19)$$

$$\beta_r = \left[\frac{\pi}{(K_r/Gr_f)}\right]\left[\frac{a_0}{2}\right] \quad (19.3\text{-}20)$$

$$a_0 = \left[\frac{2\pi r_f}{\tilde{T}v_s}\right] \quad (19.3\text{-}21)$$

Table 19.3-1 Effective Shear Wave Velocity Ratio (v_s/v_{so})

	Effective Peak Acceleration, $S_{DS}/2.5$[a]			
Site Class	$S_{DS}/2.5 = 0$	$S_{DS}/2.5 = 0.1$	$S_{DS}/2.5 = 0.4$	$S_{DS}/2.5 \ge 0.8$
A	1.00	1.00	1.00	1.00
B	1.00	1.00	0.97	0.95
C	1.00	0.97	0.87	0.77
D	1.00	0.95	0.71	0.32
E	1.00	0.77	0.22	[b]
F	[b]	[b]	[b]	[b]

[a]Use straight-line interpolation for intermediate values of $S_{DS}/2.5$.
[b]Site-specific geotechnical investigation and dynamic site response analyses shall be performed.

Table 19.3-2 Effective Shear Modulus Ratio (G/G_0)

	Effective Peak Acceleration, $S_{DS}/2.5$[a]			
Site Class	$S_{DS}/2.5 = 0$	$S_{DS}/2.5 = 0.1$	$S_{DS}/2.5 = 0.4$	$S_{DS}/2.5 \ge 0.8$
A	1.00	1.00	1.00	1.00
B	1.00	1.00	0.95	0.90
C	1.00	0.95	0.75	0.60
D	1.00	0.90	0.50	0.10
E	1.00	0.60	0.05	[b]
F	[b]	[b]	[b]	[b]

[a]Use straight-line interpolation for intermediate values of $S_{DS}/2.5$.
[b]Site-specific geotechnical investigation and dynamic site response analyses shall be performed.

$$\beta_{rr} = \left[\frac{(\pi\psi/4)a_0^2}{(K_{rr}/Gr_f^3)[2+a_0^2]}\right]\left[\frac{a_0}{2\alpha_{rr}}\right] \quad (19.3\text{-}22)$$

$$\psi = \sqrt{\frac{2(1-\nu)}{(1-2\nu)}} \le 2.5 \quad (19.3\text{-}23)$$

$$\alpha_{rr} = 1.0 - \left[\frac{0.35a_0^2}{1.0+a_0^2}\right] \quad (19.3\text{-}24)$$

where

r_f = radius of the circular foundation;
v_s = the average effective shear wave velocity over a depth of r_f below the base of the structure determined using v_{so} and Table 19.3-1 or a site-specific study;
v_{so} = the average low strain shear wave velocity over a depth of r_f below the base of the structure; and
γ = the average unit weight of the soils over a depth of r_f below the base of the structure.

19.3.5 Soil Damping. The effects of soil hysteretic damping shall be represented by the effective soil hysteretic damping ratio, β_s, determined based on a site-specific study. Alternatively, it is permitted to determine β_s in accordance with Table 19.3-3.

19.4 KINEMATIC SSI EFFECTS

Kinematic SSI effects are permitted to be represented by response spectral modification factors RRS_{bsa} for base slab averaging and RRS_e for embedment, which are multiplied by the spectral acceleration ordinates of the response spectrum at each period. The modification factors are calculated in accordance with Sections 19.4.1 and 19.4.2. Modifications of the response spectrum for kinematic SSI effects are permitted only for use with the nonlinear response history analysis provisions of Chapter 16 using the site-specific response spectrum developed in accordance with Chapter 21 and subject to the limitations in Sections 19.2.3, 19.4.1, and 19.4.2.

The product of $RRS_{bsa} \times RRS_e$ shall not be less than 0.7.

19.4.1 Base Slab Averaging. Consideration of the effects of base slab averaging through the development of site-specific transfer functions that represent the kinematic SSI effects expected at the site for a given foundation configuration is permitted.

Alternatively, modifications for base slab averaging using the procedures of this section are permitted for the following cases:

1. All structures located on Site Class C, D, or E; and
2. Structures that have structural mats or foundation elements interconnected with concrete slabs or that are continuously connected with grade beams or other foundation elements of sufficient lateral stiffness so as not to be characterized as flexible under the requirements of Section 12.3.1.3.

The *modification* factor for base slab averaging, RRS_{bsa}, shall be determined using Eq. (19.4–1) for each period required for analysis.

$$RRS_{bsa} = 0.25 + 0.75 \times \left\{\frac{1}{b_0^2}\left[1-\left(\exp\left(-2b_0^2\right)\right) \times B_{bsa}\right]\right\}^{1/2} \quad (19.4\text{-}1)$$

where

$$B_{bsa} = \begin{cases} 1 + b_0^2 + b_0^4 + \frac{b_0^6}{2} + \frac{b_0^8}{4} + \frac{b_0^{10}}{12} & b_0 \le 1 \\ [\exp(2b_0^2)] \times \left[\frac{1}{\sqrt{\pi}b_0}\left(1-\frac{1}{16b_0^2}\right)\right] & b_0 > 1 \end{cases} \quad (19.4\text{-}2)$$

$$b_0 = 0.00071 \times \left(\frac{b_e}{T}\right) \quad (19.4\text{-}3)$$

where

b_e = effective foundation size (ft),

$$b_e = \sqrt{A_{\text{base}}} \le 260 \text{ ft} \quad (19.4\text{-}4)$$

$$650 \le v_s \le 1{,}650$$

where v_s is in ft/s.

In SI units,

$$b_0 = 0.0023 \times \left(\frac{b_e}{T}\right) \quad (19.4\text{-}3.\text{si})$$

where

b_e = effective foundation size (m),

$$b_e = \sqrt{A_{\text{base}}} \le 80 \text{ m} \quad (19.4\text{-}4.\text{si})$$

$$200 \le v_s \le 500$$

where v_s is in m/s;

T = response spectra ordinate period, which shall not be taken as less than 0.20 s when used in Eq. (19.4-3) or (19.4-3.si); and
A_{base} = area of the base of the structure [ft^2 (m^2)].

19.4.2 Embedment. The response spectrum shall be developed based on a site-specific study at the depth of the base of the structure. Alternatively, modifications for embedment are permitted using the procedures of this section.

The modification factor for embedment, RRS_e, shall be determined using (Eq. 19.4-5) for each period required for analysis.

$$RRS_e = 0.25 + 0.75 \times \cos\left(\frac{2\pi e}{Tv_s}\right) \quad (19.4\text{-}5)$$

where

e = foundation embedment depth [ft (m)], not greater than 20 ft (6.1 m). A minimum of 75% of the foundation footprint shall be present at the embedment depth. The foundation

Table 19.3-3 Soil Hysteretic Damping Ratio, β_s

	Effective Peak Acceleration, $S_{DS}/2.5$[a]			
Site Class	$S_{DS}/2.5 = 0$	$S_{DS}/2.5 = 0.1$	$S_{DS}/2.5 = 0.4$	$S_{DS}/2.5 \ge 0.8$
C	0.01	0.01	0.03	0.05
D	0.01	0.02	0.07	0.15
E	0.01	0.05	0.20	[b]
F	[b]	[b]	[b]	[b]

[a]Use straight-line interpolation for intermediate values of $S_{DS}/2.5$.
[b]Site-specific geotechnical investigation and dynamic site response analyses shall be performed.

embedment for structures located on sloping sites shall be the shallowest embedment;

v_s = the average effective shear wave velocity for site soil conditions, taken as average value of velocity over the embedment depth of the foundation determined using v_{so} and Table 19.3-1 or a site-specific study and shall not be less than 650 ft/s (200 m/s);

v_{so} = the average low strain shear wave velocity over the embedment depth of the foundation; and

T = response spectra ordinate period, which shall not be taken as less than 0.20 s when used in Eq. (19.4-5).

19.5 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

See Chapter 23 for the list of consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.